An Improved Neural Network Classification Algorithm by Expanding Training Samples for Polarimetric SAR Application

Jing Chen, Biao Hou, *Member, IEEE*, Bo Ren, *Member, IEEE*, Qian Wu, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— The idea of spatial correlation has been used in polarimetric synthetic aperture radar (PolSAR) classification for many years. It is common that the bigger the spatial correlation, the more information it contains. Though the recent advances in deep learning for PolSAR classification have achieved remarkable progress, the number of training samples is still a problem that we have to face. Aiming to solve this problem, it is valuable to explore a greater spatial correlation of terrains as a priori for classification. Considering the correlated regions of the training sample as a prior, rather than a small neighborhood, this article proposed the Wishart locally constrained expansion (WLCE) algorithm. Based on this, a PolSAR classification algorithm is designed. The whole proposed PolSAR classification algorithm includes three parts: the WLCE algorithm, the convolution neural network (NN) training algorithm, and the Markov random field (MRF) postprocessing algorithm. Supervised by the cluster knowledge from the Wishart classifier, the WLCE algorithm expands samples iteratively from the regions related to the training samples with a region-expanding technique, where very few labeled samples turn into a larger training set. The convolution NN (CNN) algorithm is designed to train CNN with three types of training samples for a more accurate result. CNN is trained first with the numerous training samples generated through the WLCE algorithm, and then, the samples from consistency extraction and raw samples are involved to fine-tune the CNN to get the improved result. Finally, MRF prior is used to smooth the result. Several benchmark datasets are adopted to evaluate the effectiveness of the proposed algorithm. The experimental results show that the new semisupervised classification (SSC) algorithm outperforms the state-of-the-art SSC algorithms and classical supervised classification algorithms.**

***Index Terms*— Polarimetric synthetic aperture radar (PolSAR), region prior, Wishart locally constrained expansion (WLCE).**

Manuscript received 5 May 2023; revised 29 July 2023; accepted 7 August 2023. Date of publication 10 August 2023; date of current version 24 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62171347, Grant 61877066, Grant 61771379, Grant 62001355, and Grant 62101405; in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61621005; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-05 and Grant 2021ZDLGY02-08; in part by the Science and Technology Program in Xi'an, China, under Grant XA2020-RGZNTJ-0021; and in part by 111 Project. *(Corresponding author: Biao Hou.)*

Jing Chen is with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Xidian University, Xi'an 710071, China, and also with the School of Physics and Electronics, Henan University, Kaifeng 475004, China (e-mail: chenjingrr@163.com).

Biao Hou, Bo Ren, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: avcodec@hotmail.com; asdwer2046@126.com; lchjiao@mail.xidian.edu.cn).

Qian Wu is with the Institute of Information and Navigation, Air Force Engineering University, Xi'an 710038, China (e-mail: zhuangzaierhao@163.com).

Digital Object Identifier 10.1109/TGRS.2023.3304115

I. Introduction

DUE to the excellent characteristics for all-weather and day/night operation [1], polarimetric synthetic aperture radar (PolSAR) has been widely used in various fields, such as terrain classification, environment monitoring, disaster management, agriculture and water resource management, and urban planning [2], [3], [4], [5]. Terrain classification plays an important role in PolSAR application. Many studies have been carried out, and lots of achievements have been made [6], [7], [8], [9], [10], [11], [12]. The studies on PolSAR image classification are traditionally classified into three categories: polarization feature-based [13], [14], [15], statistical distribution-based [6], [7], [8], [9], [10], and machine learning-based [16], [17].

Polarization feature-based methods that are related to the physical properties of terrains focus on identifying different terrain by polarimetric decomposition. In this view, a series of polarization [13], [15], [18], [19] methods have been put forward to classify the PolSAR image. These features are usually directly sent to a classifier or used to obtain high-level features (for example, wavelet feature, Gabor feature, and sparse representation) and then sent to certain classifiers, which can be support vector machine (SVM), spectral clustering, ensemble learning, and so on.

Some researchers employed various types of statistical distribution of PolSAR data to solve the problem of classification [6], [7], [8], [9], [10]. Most known of all, the polarimetric covariance matrix and the coherency matrix of data are assumed to be subject to a complex Wishart distribution [20]. In order to describe heterogeneous regions, a product model is usually used to add texture information. According to different texture modeling in the product model, various distributions [21], [22], [23] have been proposed to model the statistical characteristics of PolSAR data. By introducing the spatial relation, some extended statistical distribution models [24], [25], [26], [27], [28] have been also proposed.

Recently, machine learning has become more and more popular in PolSAR image classification. Up until now, many PolSAR image classification methods are proposed

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

based on machine learning, such as expectation maximization (EM) algorithm [16], tritraining [29], boosting [17], neural networks (NNs) [11], [30], [31], [32], [33], [34], [35], random forest classifier [36], and SVM [37]. With the booming development of deep learning in recent years, many deep learning classification methods have been put forward, such as polarimetric-feature-driven deep convolution NN (CNN) [11], Wishart DSN [12], Wishart-AE and Wishart-CAE [38], MAE [39], VAE to train Wishart classifier parameter [40], superpixel-driven optimized Wishart network that combines spatial information of superpixels [41], mutual information-based CNN [42], long short-term memory (LSTM) [43], convolution LSTM [44], deep Q-network (DQN) [45], stack sparse autoencoder using local spatial information [46], and random neighbor pixel-block-based deep recurrent (DR) learning [47]. Besides, CNN has been proposed for PolSAR image classification, such as CV-CNN [48], shallow CNN [49], 3-D CNN [50], and CV-ConvLSTM [3].

Most of the above classification methods are supervised learning. Supervised learning generally gets better classification performance than unsupervised learning. However, without enough training samples, the classification is poor, whereas, due to the complex imaging mechanism, obtaining sufficient labeled samples where expert knowledge is required is not only time-consuming but also labor-intensive. Therefore, a number of semisupervised classification (SSC) methods that use both labeled and unlabeled samples in training have been proposed. Geng et al. [51] proposed a superpixel-restrained deep NN with multiple decisions (SRDNN-MD) for PolSAR image classification. Liu et al. [52] constructed a semisupervised deep metric learning network for the classification of PolSAR data that introduces a new manifold regularization inspired by distance metric learning. Fang et al. [53] put forward a semisupervised method that uses K-means pseudolabels in a 3-D-CNN model to perform the classification task. Wang et al. [29] proposed a tritraining SSC algorithm that uses MST pseudolabels' expansion. Bi et al. [50] propose a graph-based semisupervised model that uses k-wishart pseudolabels or superpixels pseudolabels to enlarge the training set and Markov random field (MRF) for PolSAR classification. Li et al. [54] put forward a semisupervised PolSAR image classification method based on superpixels. Hua et al. [55] proposed a semisupervised cotraining algorithm for PolSAR image classification that uses superpixel to enlarge the training set. Wen et al. [48] proposed an SSC algorithm that uses Wishart pseudolabels to enlarge the training set for CV-CNN. Bi et al. [56] integrated active learning and CNN into a principled framework.

Though some satisfying results have been achieved in PolSAR classification, there are still some unsolved problems. Many current semisupervised PolSAR classification algorithms often need more labeled, which usually has a labeled sample rate of no less than 1%. The number and reliability of pseudolabels still remained to be strengthened. For example, the simple clustering algorithm (k-means, Wishart, and so on) cannot be limited by number, but the reliability of pseudolabels is low. The superpixel-based algorithm is affected by the superpixel parameters. The reliability of the label is difficult to guarantee if the superpixel is too large, and the samples generated are too few if the superpixel is too small.

A. Contribution

In this article, a novel PolSAR classification algorithm is proposed to classify PolSAR data, which we call WCCNM. WC stands for Wishart locally constrained expansion (WLCE), CN stands for CNN, and M stands for MRF. This algorithm uses very few labeled samples for the PolSAR image classification task.

Our classification algorithm focuses on expanding training samples. The premise of excellent expansion of the training samples should be excellent prior. Whether it is statistical distribution-based, polarization feature-based, or machine learning-based, the algorithm of a single pixel is not ideal. Considering the spatial correlation [46] of pixels is very important for image classification. Superpixel technology [51], MRF technology, and traditional window feature extraction technology (for example, window-based texture feature and CNN's patch) are all successful applications of pixel spatial correlation in image classification.

However, the spatial correlation characteristic considered previously is limited. In fact, the PolSAR image is composed of multiple terrain regions, where a pixel of a specific category hardly appears separately. Each pixel belongs to a specific category-related region, which is always larger than the previously considered region. It is common that the bigger the spatial correlation region, the more information it contains. It is valuable to explore a greater spatial correlation of terrains as a priori for classification. In order to facilitate the description, the spatial correlation that we aim to explore is referred to as sample-related region prior here. The cluster assumption is an important idea for semisupervised learning, which assumes that the patterns in the same cluster are likely to be of the same class. As a classical statistical distribution, Wishart can directly describe the statistical characteristics of the original PolSAR data, which needs only a small number of training samples. The successful application of Wishart prior in deep learning also confirms its vitality [12], [38], [39], [40].

This article takes the sample-related region prior and the information of the Wishart cluster into consideration to generate unlabeled samples. Starting from a traditional Wishart classifier, this article iteratively obtains enough effective labels within the constrained local field with very few labeled pixels.

There is no extensive consideration of classifiers, and WCCNM uses CNN to perform classification. CNN is an excellent classifier and has good fault tolerance, but it is often a hard task to obtain good results with limited training samples. WCCNM uses current limited labeled data to expand a reliable training set through Wishart distribution for CNN, trying to make full use of the advantages of Wishart and CNN, and avoid their disadvantages. When the expanded training samples are obtained, they are fed into CNN for primary training. Then, the samples from consistency extraction and raw samples are also used to fine-tune the CNN to get an improved result. After CNN is trained, the result is postprocessed by MRF to get a more context-related result.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

The main contributions of this article are summarized as follows.

1) A bigger spatial prior is proposed. A sample is associated with a region, rather than just a small neighborhood.
2) The WLCE algorithm is proposed to expand effective and enormous training samples based on Wishart statistical distribution and the region prior associated with the training samples.
3) Two WLCE rules are proposed and experimented with. The WLCE algorithm uses the Wishart classifier to expand very few labeled samples to a larger training set while taking into account the inhomogeneity of images. The comparison between the two WLCE rules gives some hints for future sample expansion design.
4) An improved two-stage training NN classification algorithm is designed to better utilize WLCE training samples.

The remainder of this article is organized as follows. First, preliminaries are introduced in Section II. Section III explains the proposed method. Section IV shows the experimental results and discussion. Finally, the conclusions are presented in Section V.

II. Preliminaries

A. Notations

PolSAR data are defined as $X = \{x_i\} \in R^{N\times D}$, and corresponding label data are defined as $Y = \{y_i \mid i \in \{1,\ldots,N\}, y_i \in \{1,\ldots,K\}\}$. Here, $N = H \times W$ is the total number of pixels, where H is the height, W is the width of the PolSAR image, and D is the dimension of the PolSAR data. The task is to assign label y_i to each pixel i ($i \in \{1,\ldots,N\}$). The labeled training samples are denoted as $\text{TrSet} = \{\text{TrSet}\{i\} \mid i \in \{1,\ldots,K\}\}$ and $\text{TrSet}\{j\} = \{(x_l, y_l) \mid y_l = j, x_l \in X\}$, $j \in \{1,\ldots,K\}$. K is the class number. M is the number of samples in TrSet. V_i is the feature of pixel i. Nei(i) is the neighborhood of pixel i. The above are some commonly used symbols in this article. Other symbols used locally are described in the corresponding sections.

B. Wishart Distribution

It has been one of the most outstanding achievements in the usage of the statistical distribution of PolSAR data. For a reciprocal target illuminated by a monostatic SAR, the polarimetric scattering information can be represented by a complex vector [2]

$$k = 1/\sqrt{2}\left[S_{xx} + S_{yy}, S_{xx} - S_{yy}, S_{xy} + S_{yx}\right]^T \tag{1}$$

where S_{ij} denotes the complex scattering coefficient with j transmitting and i receiving antennas, $S_{xy} = S_{yx}$, and superscript T denotes the matrix transpose. For multilook PolSAR data, the polarimetric information can be represented by an average coherence matrix

$$T = 1/L \sum_{i=1}^{L} k_i k_i^H \tag{2}$$

where superscript H denotes the complex conjugate and L is the number of looks. Let $Z = LT$. Matrix Z has a complex Wishart distribution [2]. The probability density function (pdf) of T is

$$W_p(T|L,\Sigma) = \frac{L^{Ld}|T|^{L-d}exp\left(-LTr\left(\Sigma^{-1}T\right)\right)}{|\Sigma|^L K(L,d)} \tag{3}$$

where d is the dimension, Tr(.) denotes the trace operation, $|\,.\,|$ denotes the determinant, and $K(L,d) = \pi^{d(d-1)/2}\prod_{i=1}^{d}\Gamma(L-i+1)$. $\Gamma(.)$ is the Gamma function.

With the Bayesian maximum likelihood method and some algorithm processing, the class probability expression mentioned above is transformed into the following measure [6]:

$$d(T|\Sigma) = ln|\Sigma| + \text{Tr}\left(\Sigma^{-1}T\right). \tag{4}$$

Traditionally, for its intractability, parameter Σ is computed as the mean value of training samples within each class. The category is determined by the following formula:

$$i = \arg\max_k W_p(T|L,\Sigma_k) = \arg\min_k d(T|\Sigma_k). \tag{5}$$

C. Convolution Neural Network for Classification

The CNN network is good at maintaining image local structure and is very useful to recognize different patterns.

1) Convolution Neural Network Layers: A CNN network mainly consists of a stack of alternating convolution layers, pooling layers, and fully connected layers. The convolution layer is the most important part of CNN for feature extraction. Usually, after the convolution layer, there exists a pooling layer that is designed to perform some local nonlinear operation on the feature maps. The pooling operation makes the representation of the extracted convolution layer features more abstract. Fully connected layers are typically the last few layers that the previous feature maps are flattened and fed in to prepare for the final decision.

2) Softmax Classifier: The softmax function is used as the final output layer to solve the multiclass classification problem. Let α_i be the output of the fully connected layers and W_s be the weight matrices of the softmax function; the output of the softmax function is

$$P(y_i = j \mid \alpha_i; W_s) = \frac{e^{W_{sj}^T\alpha_i}}{\sum_{k=1}^{K} e^{W_{sk}^T\alpha_i}}. \tag{6}$$

3) Loss Function Optimization: The following loss function can be used to perform the CNN classification task. θ represents the parameters of CNN

$$\text{Loss} = -\sum_{i=1}^{M}\sum_{k=1}^{K} I\{y_i = k\}\log\ P(y_i = k \mid \theta). \tag{7}$$

D. Markov Random Field

Whether it is statistical distribution-based, polarization feature-based, or machine learning-based, the algorithm of a single pixel is not ideal, and adding spatial correlation can improve the classification performance greatly. MRF, which can be used to encourage the label to be consistent with its neighbors, is skillful in incorporating the spatial prior of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

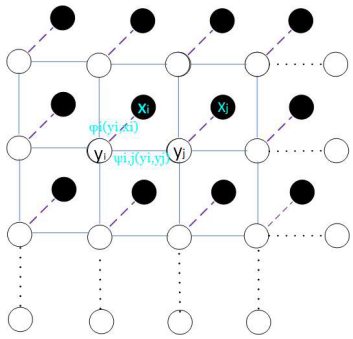


Fig. 1. MRF.

images and has been widely applied to remote sensing image classification [24], [57]. MRF can be solved by minimizing energy function E, which is composed of two parts. One part represents unary data energy E_d, and the other part represents pairwise spatial smooth energy E_s

$$E = E_d + E_s. \tag{8}$$

E_d is data energy and E_s is smooth energy. A more specific form is given as

$$E = -\sum_{i=1}^{N} \log P(x_i|\Theta) + \alpha_s \sum_{i=1}^{N} \sum_{j\in \mathrm{Nei}(i)} \left(1 - I\left(y_i - y_j\right)\right) \exp\left(-\frac{\|v_i - v_j\|_2^2}{2\sigma}\right). \tag{9}$$

Here, α_s is the pairwise smooth coefficient. $I(.)$ is the indicator function, and v_i is the feature vector of pixel i, which is set to be the same as x_i in this article. σ is the mean squared distance between features of adjacent pixels i and j.

The pairwise smoothness term enforces that the adjacent pixels that have similar values should take the same label. If pixels i and j are similar, $\exp(-(\|v_i - v_j\|_2^2/2\sigma))$ in (9) is large; then, minimizing E strongly enforces that labels y_i and y_j should have the same value [58].

An MRF example is shown in Fig. 1. Label y_i is linked to the feature data x_i or v_i (they can be the same). $\phi_i(y_i, x_i)$ is the likelihood potential of label y_i given feature data x_i, taking the form of the multiclass probabilistic form $-\log P(y_i = j|x_i)$ from some probabilistic classifier, which is the E_d part of MRF. $\psi_{i,j}(y_i, y_j)$ is the label smoothness potential that encourages label contextual consistency between neighboring pixels i and j, which is the Es part of MRF.

III. Proposed Algorithm

The complete framework of the proposed WCCNM algorithm is shown in Fig. 2. The algorithm consists of three parts: the WLCE algorithm, the CNN training algorithm, and the MRF algorithm. First, the WLCE algorithm iteratively expands the original training samples to a large training set supervised by Wishart prediction in the sample-related region. Then, CNN is trained with RF feature data, which is a 6-D feature matrix. There are two stages in the CNN training process. In the first stage, CNN uses the samples expanded by the WLCE algorithm for training to obtain preliminary results. In the second stage, CNN is fine-tuned through the consistency samples and the original training samples to obtain further results. The consistency samples are those samples with consistency labels between WLCE labels and the preliminary CNN prediction results. The original training sample set intervenes under certain conditions in the CNN fine-tuning as distribution correction in fine-tuning. Finally, MRF is used to smooth the results. More details can be obtained in the subsequent sections, which are further shown in Algorithms 1 and 2.

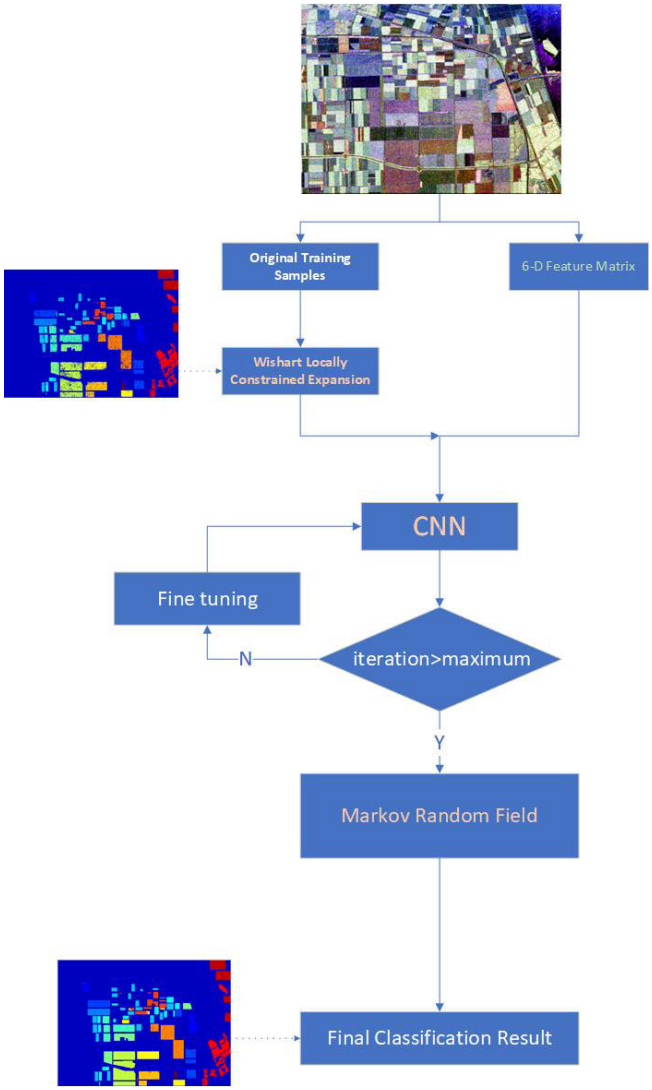


Fig. 2. Framework of the proposed PolSAR classification algorithm.

A. Wishart Locally Constrained Expansion

Each PolSAR image is composed of multiple terrain regions, where there is a pixel, and there is a specific category region associated with it. It is reasonable to expand the pixels in the region associated with a given training sample as training pseudosamples. The cluster assumption is an important idea for semisupervised learning, which assumes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

that the patterns in the same cluster are likely to be of the same class. This article tries to combine the two ideas for samples' expansion. Supervised by the cluster knowledge from the Wishart classifier, the WLCE algorithm expands samples iteratively from the regions associated with the training samples, where very few labeled samples turn into a larger training set. We designed an iterative algorithm based on each round of sample expansion. It can be seen as a kind of supervised region-growing algorithm. Different from the traditional region growing algorithm that is usually supervised by a given threshold or relative to the immediate neighbor information, each growing step in the proposed algorithm is supervised by the Wishart classifier and loose neighbor information (local constraint).

Like the traditional region growing, the first step is to determine extension rules. Here, two different growing rules are proposed, which are the cluster center rule and the sample center rule described in Rule 1 and Rule 2.

Rule 1 (Cluster-Centered Growing Rule): For a training sample T' with a known label K, if there is a pixel T in its neighborhood classified as class K by the Wishart classifier, the pixel T is expanded into class K training set. Here, $K = \arg\max_i W_p(Tc|L, \Sigma_i)$.

Rule 2 (Sample-Centered Growing Rule): For a training sample T' with a known label K, if certain pixel T in its neighborhood and the training sample T' are at the same time classified as class K' by the Wishart classifier, the pixel T in its neighborhood is expanded into class K training set. Here, $K' = \arg\max_i W_p(Tc|L, \Sigma_i) = \arg\max_i W_p(Tc'|L, \Sigma_i)$, and $K = \text{TrLabel}(T')$.

In order to describe the expansion algorithm more exactly, some equations are introduced as follows:

$$g = I\left(T' \in \text{TrSet}, \text{TrLabel}\left(T'\right) = K, T \in \text{Nei}\left(T'\right)\right) \tag{10}$$

$$g1 = I\left(\arg\max_i W_p(Tc|L, \Sigma_i) = K\right) \tag{11}$$

$$g2 = I\left(\arg\max_i W_p(Tc|L, \Sigma_i) = \arg\max_i W_p\left(Tc'|L, \Sigma_i\right)\right) \tag{12}$$

$$f\left(T, T', \Sigma_i\right) = g\left(T, T'\right) gu (gu = g1 \text{ or } g2) \tag{13}$$

$$k = Kf\left(T, T', \Sigma_i\right). \tag{14}$$

The function $g(.)$ indicates the local area constraint. Two functions $g1$ and $g2$ correspond to two rules: cluster-centered growing rule and sample-centered growing rule.

TrSet is the set of the labeled training samples. TrLabel(i) is the label of pixel i. T' is the currently investigated training pixel sample with coherent matrix Tc'. T is the currently investigated neighbor pixel sample with coherent matrix Tc. Nei(T') is a W square neighbor region around the pixel T'. $I(.)$ indicates the indicator function. $T \in \text{Nei}(T')$ indicates the candidate of the locally constraint expansion. The functions $g1(.)$ and $g2(.)$ indicate the Wishart constraint of the expansion.

The values of g are designed as 0 and 1 to fit for sample expansion. If g equals 0, sample T is not selected for sample expansion. If g is above 0, K is the sample expansion label for sample T. In total, (14) shows the principle of our WLCE.

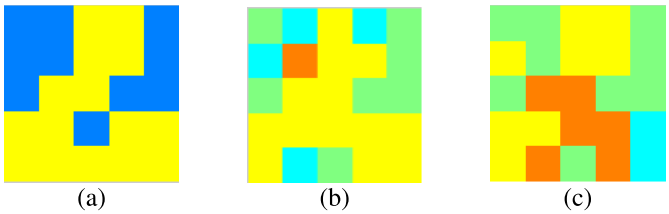


Fig. 3. WLCE rule. (a) Ground-truth patch, a 5×5 patch from ground-truth map. (b) Case 1: the central pixel is correctly classified, and most of the neighbor pixels are correctly classified; both Rule 1 and Rule 2 expand the yellow region. (c) Case 2: orange indicates the classified class of the central pixel, and the central pixel is not correctly classified; Rule 1 expands the yellow region, and Rule 2 expands the orange region.

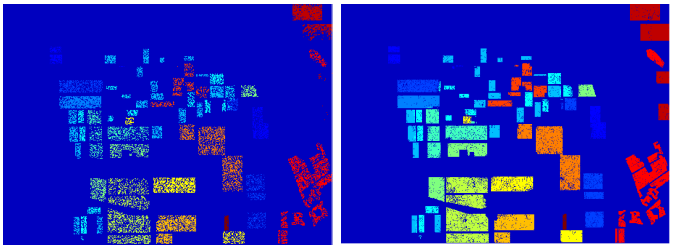


Fig. 4. Flevoland sample expansion with WLCE rule. (a) Expansion result with Rule 1. (b) Expansion result with Rule 2.

The cluster-centered growing rule only grows the pixels that are around the currently investigated training sample and obtained the same Wishart prediction label as the training sample's ground-truth label (function $g1(.)$, centering on cluster label). It is inspired by self-learning, which searches for the high confidence samples in the prediction result. Here, the confidence comes from the local area constraint of certain training samples. As shown in some studies, clustering results can serve as pseudolabels to improve algorithm performance [48], [50], [53]. Wishart prediction labels themselves can be used as pseudolabels for semisupervised learning. Rule 1 affords an extra guarantee [function $g(.)$] for the pseudolabels.

The idea of Rule 1 is shown in Fig. 3. Constrained by $g1$, the yellow area is expanded in cases 1 and 2. The Wishart prediction label of the extension samples should be yellow (cluster-centered).

The sample-centered growing rule grows the pixels that are around the currently investigated training sample and obtained the same Wishart prediction label from the Wishart classifier as what the currently investigated training sample obtains from the Wishart classifier (function $g2(.)$, centering on sample prediction label), where it is not necessarily that the samples have the same Wishart prediction label as the currently investigated training sample's ground-truth label.

This idea is based on the fact that, when the data feature fails to fully represent the data distribution, the classification results are often inaccurate, which is very common in classification. For the data feature with strong noise or without considering neighborhood information, the naive Wishart classification result is usually discontinuous in the inhomogeneous region, where samples belonging to the same category will be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

classified into a certain same false category that is inconsistent with the real category. This shows that, for an unlabeled pixel sample, if there are labeled samples nearby, the Wishart classification result is usually more inaccurate than directly setting the classification result to the same label as the training label nearby under the circumstances. This article assumes that the adjacent pixels of the supervised samples from different classes have a poor chance that they are classified as a certain same class by the Wishart classifier. Once the pixels that are around the currently investigated training sample obtain the same Wishart prediction label from the Wishart classifier as what the currently investigated training sample obtains from the Wishart classifier, the pixel labels are set the same as the ground-truth label of the currently investigated training sample. Given specific constraints, this setting will have credibility. Here, the constraint is from Wishart classification and local constraint.

Although Rule 1 can also expand inhomogeneous samples, Rule 2 considers the existence of inhomogeneity more purposefully, which is very important in classification. The idea of Rule 2 is shown in Fig. 3. Constrained by $g2$, the yellow area is expanded in case 1, and the orange area is expanded in case 2. The Wishart prediction label of the extension samples should be yellow in case 1 (sample-centered). The Wishart prediction label of the extension samples should be orange in case 2 (sample-centered).

The candidate samples surrounding training samples can be expanded according to (14). However, it is not the end of sample expansion. If we only expand new pixels around the training sample pixels, the expansion number is limited. Meanwhile, we boldly assume that this credibility can be passed on. In this way, the training samples are continuously expanded in the neighborhood until the expansion converges or reaches the maximum number of iterations.

Considering the prediction discontinuity caused by inhomogeneity, the size of w is usually set to be more than 3 (seed pixel as center), or semiwindow (center pixel excluded) is set to more than 1. It is not proper to set semiwindow sizes too small or too large. Too small is not enough for inhomogeneity, and too large is easy to include error data. Sample-centered growing is intended to add more inhomogeneous information.

The final expansion algorithm is designed as iterative expansion, which is shown in Algorithm 1. This algorithm considers label overlapping.

The living examples of the algorithm using the two rules are shown in Fig. 4.

Algorithm 1 Wishart Locally Constrained Expansion

input: T Matrix ($m \times n \times 9$), MaxIter, Training Samples, Neighbor size w, Rulesetting
output: WLCE Training samples

initialization
neighbor window size $width = height = w$, unselect pixel samples set Sups
ite=0
While $ite < MaxIter$
  Initialize Removeset to be empty
  Use labeled pixel samples to construct Wishart classifier $WLCE^{ite}$
Use $WLCE^{ite}$ to get new label nL^{ite}
  For each training sample x_i do
    $K = TrLabel(x_i)$
    Find each unlabeled pixel x_j in the $w \times w$ neighborhood region around x_i
    Switch Rulesetting
      1:If $nL^{ite}(x_j) = K$ do
        Mark label of x_j as K
        Add x_j to Training samples
        Put index of x_j into Removeset
      End if
      2: If $nL^{ite}(x_j) = nL^{ite}(x_i)$ do
        Mark label of x_j as K
        Add x_j to Training samples
        Put index of x_j into Removeset
      End if
    End switch
End for
Remove samples in Removeset from Sups
End while
Output: WLCE Training samples

B. Convolution Neural Network

WLCE training samples obtained above altogether with the original training samples are fed into the CNN classifier to get the classification result. The architecture of the CNN designed for classification inherits from the successful architecture in [58] with a little change. The changed architecture is shown in Fig. 5. The data patches put into CNN are $8 \times 8 \times 6$ data patches, which are the same as those in [58]. The CNN includes three convolutional layers with kernel sizes of 3×3, 3×3, and 2×2, respectively, kernel numbers of 30, 20, and 20, respectively, and two max-pooling layers with pooling size set as 2×2; the stride is set as 2; and two fully connected layers contain 500 units and K units, respectively.

The CNN training process can be found in Algorithm 2. RF is a 6-D feature matrix calculated from matrix T, which is shown as follows:

$$\begin{aligned} \mathrm{RF} = \big[&10\log(\mathrm{SPAN}), T22/\mathrm{SPAN}, T33/\mathrm{SPAN} \\ &|T12|/\sqrt{T11.T22}, |T13|/\sqrt{T11.T33}, \\ &|T23|/\sqrt{T22.T33}\big] \end{aligned}$$
$$\text{where } \mathrm{SPAN} = T11 + T22 + T33. \tag{15}$$

There are two stages in the CNN training process. In the first stage, WLCE samples are used to train CNN to obtain preliminary results. In the second stage, the CNN is fine-tuned for further results.

With WLCE samples, in most cases, excellent classification results can be obtained through CNN training, but, if the sample overlapping is severe, sometimes, the results may have small deviations. To address this issue, this article introduced fine-tuning into CNN training. There are two types of fine-tuning. The first type of fine-tuning is sample consistency fine-tuning, which trains CNN through the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

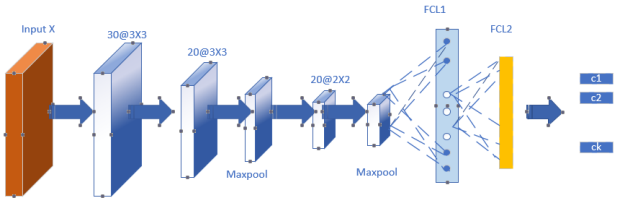


Fig. 5. Architecture of CNN.

samples with consistency labels between WLCE samples and prediction results. Sample consistency constraint removes those overlapping samples expanded in the WLCE stage, allowing CNN to use more accurate samples for fine-tuning. The second type of fine-tuning is original sample intervention fine-tuning, which assumes that the distribution learned by the CNN model based on WLCE samples fits the true distribution well except for a small deviation. By intervening in the original samples fine-tuning, it can compensate for the information loss caused by training samples generated from the region prior. It is noted here that, if the CNN model trained with WLCE samples predicts the original samples with high accuracy, then the trained model is considered to be close to the true distribution, and fine-tuning through the original samples can be omitted. During the training process, fine-tuning criteria can be set according to specific circumstances.

Algorithm 2 CNN training and MRF postprocessing in WCCNM algorithm

input:PolSAR data RF, Training Samples $WLCETS$ obtained from Algorithm 1, Maximum of iteration parameter $MaxI$, Original training samples intervenention criterion Sic
output: classification result

CNN initialization
Use samples $WLCETS$ to train CNN to get the full classification map FCM of the PolSAR image
While $iter<MaxI$
  Extract identical samples ITS between FCM and WLCE map
  Evaluate intervenention criterion Sic according to FCM
  If Sic is True
    $TS1$= Original training samples
else
    $TS1$ ={}
End If
  Use $TS1$ to fine tune CNN and get new full classification map FCM
  Use TS to fine tune CNN and get new full classification map FCM
End while
Perform MRF process of the CNN results with minimal error in labeled data
Output: classification result

C. Markov Random Field

Solving an MRF problem is NP-hard. Approximate optimal solutions can be obtained by using optimization algorithms, such as ICM, graph cut, and belief propagation (BP).

Among various optimization algorithms, the BP algorithm is a good choice.

To solve MRF using the BP algorithm, all messages are initialized as 0. The BP algorithm passes messages $\phi_i(y_i, x_i)$ and $\psi_{i,j}(y_i, x_i)$ iteratively in an *up–down–left–right* way in MRF to get the optimization result. At iteration t, the message m_{ij}^t sent from node i to its neighbor j is given as follows:

$$m_{ij}^t(y_j) \leftarrow \max_{y_i}\left(\sum_{k\in \mathrm{Nei}(i)\setminus j} m_{ki}^{t-1}(y_i) - \phi_i(x_i, y_i) - \psi_{i,j}(y_i, y_j)\right). \tag{16}$$

The messages of each node are iteratively updated and propagated to its neighboring nodes until convergence. Finally, the class labels are obtained by estimating the maximum belief for each node i based on its incoming messages, which is given by

$$y_i = \arg\max_{y_i\in\{1,\ldots,K\}}\left(\sum_{k\in \mathrm{Nei}(i)} m_{ki}^t(y_i) - \phi_i(x_i, y_i)\right). \tag{17}$$

IV. Experiments

In this part, we evaluate our algorithm on four real benchmark PolSAR datasets from different perspectives.

A. Experiment Datasets and Setting

CNN architecture has been shown in Section III. The data input into CNN are simple 6-D PolSAR data as in [58]. For CNN optimization, the batch size is set as 100. The learning rate is set to 0.001, and the momentum parameter is 0.9. The iteration number for MRF is set as 10. 0.5% labeled pixels are randomly selected as training samples.

There are four benchmark PolSAR datasets in our experiments: San Francisco Bay data, Oberpfaffenhofen data, Flevoland data, and Weihe River data. Fig. 6 shows the Pauli images and the ground-truth maps of these datasets. On the left are Pauli images, and on the right are the ground-truth maps. The unlabeled regions are the regions without consideration in the experiments.

San Francisco Bay data are a C-band 25-look PolSAR dataset for San Francisco Bay, which was acquired by RADARSAT-2 in 2008. The size is 1380 × 1800. The resolution of the image is 8 × 8 m. There are five types of terrains in the image: water, vegetation, low-density urban, high-density urban, and developed.

Oberpfaffenhofen data are an L-band PolSAR dataset for the Oberpfaffenhofen area in Germany, which was obtained by E-SAR, German Aerospace Center. The size of this image is 1300 × 1200. There are three types of terrains under consideration in the image: built-up areas, wood land, and open areas.

Flevoland data are an L-band four-look PolSAR dataset for the Flevoland area in The Netherlands, which was acquired by the National Aeronautics and Space Administration/Jet Propulsion Laboratory (NASA/JPL) AIRSAR on August 16, 1989. The size of the image is 750 × 1024. The resolution of the image is 12 × 6 m. There are 15 classes of crops in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

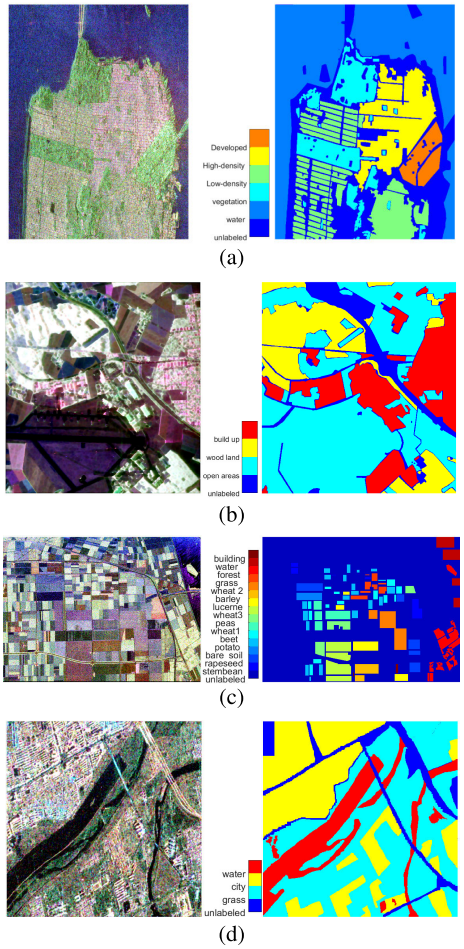


Fig. 6. PolSAR data. (a) San Francisco Bay data. (b) Oberpfaffenhofen data. (c) Flevoland data. (d) Weihe River data.

image: water, barley, peas, stembean, lucerne, building, beet, rapeseed, potato, forest, bare, soil, grass, wheat 1, wheat 2, and wheat 3.

Weihe River data are a C-band single-look PolSAR dataset for Weihe River in Xi'an, which was acquired by RADARSAT-2 in 2010. The size is 512 × 512. The resolution of the image is 8 × 8 m. There are three types of terrains in the image: water, grass, and city.

To simplify the experiments without reduction of the evaluation for the proposed algorithm, the experiments are designed as follows. The San Francisco Bay and Oberpfaffenhofen datasets are used to show the effectiveness of WLCE sample expansion. The Flevoland and Weihe River datasets—the former stands for more classes and detailed, and the latter is the dataset with more confusion result in the current literature, which is typically difficult to be classified in some terrains—are used to show the performance following the overall process of the proposed algorithm. Besides, all the datasets are denoised using the refined Lee filter.

B. Results for Wishart Locally Constrained Expansion

This article presents a very effective sample expansion algorithm.

First, an overview is used to show the efficiency of the proposed WLCE algorithm. Here, eight groups of WLCE experiments have been done on San Francisco Bay data and Oberpfaffenhofen Data.

WLCE has three parameters that are expansion window size, expansion iteration number, and rule option that indicates cluster-centered expansion or sample-centered expansion. Eight groups of data correspond to two rules and two typical sizes of two data. The reason why the iteration number is set as 8 is that the result of the different number of iterations can be obtained in the whole iteration process, and the algorithm of iteration number 8 is close to converge or making the image approximate full-covering state. WLCE uses 0.5% of the ground-truth pixels for the initial state.

Figs. 7 and 8 show eight groups of WLCE iterations. Fig. 7 shows the WLCE iterations of San Francisco Data. Fig. 8 shows the WLCE iterations of Oberpfaffenhofen Data. Figs. 7(a) and (b) and 8(a) and (b) show the WLCE iterations with expansion window size setting as 7. Figs. 7(c) and (d) and 8(c) and (d) show the WLCE iterations with expansion window size setting as 11. Figs. 7(a) and (c) and 8(a) and (c) show the WLCE iterations with the cluster-centered rule. Figs. 7(b) and (d) and 8(b) and (d) show the WLCE iterations with the sample-centered rule. We can observe that, neither for San Francisco Bay data nor Oberpfaffenhofen data, the expansion is effective. As the iterations go on, more and more unlabeled samples are expanded. Obviously, Rule 1 usually grows fewer samples than Rule 2. WLCE can expand many samples, which is useful to train CNN. In particular, for the sample-centered rule, the growing samples are about to cover the whole image. Compared with ground truth, the expanded samples have clear category boundaries and high accuracy. This is very useful for obtaining classification results with high accuracy. From the figures, we can see that, for the expansion of the sample-centered rule, the final results have surpassed many existing classification algorithms.

In fact, except for the last iteration of Fig. 7(c), the accuracy of nonoverlapping samples exceeds 99%. The worst case is the accuracy of the last iteration of Fig. 7(c), which is still above 98%, while the accuracy of nonoverlapping samples in Fig. 8 exceeds 97%. The coverage of nonoverlapping samples in the image for eight iterations exceeds 93%. The coverage of nonoverlapping samples on the image for eight iterations exceeds 93%, Figs. 7 and 8 fully demonstrate the effectiveness of the WLCE algorithm.

The WLCE results of the Flevoland and Weihe River datasets are shown in Figs. 9 and 10. In order to display detailed information and reduce redundant display, we show the results of the WLCE algorithm on the two datasets with four iterations. The WLCE samples are expanded according to Rule 1 and Rule 2, with window sizes of 7 and 11.

The gray part in Figs. 7–10 represents samples with overlapping categories. It can be seen that the number of overlapping samples is very limited in three datasets except for the Weihe River dataset. In Figs. 7 and 8, the gray part is almost invisible.

We can see that, except for the Weihe River dataset with obvious sample overlapping, all other results are close to ground truth.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

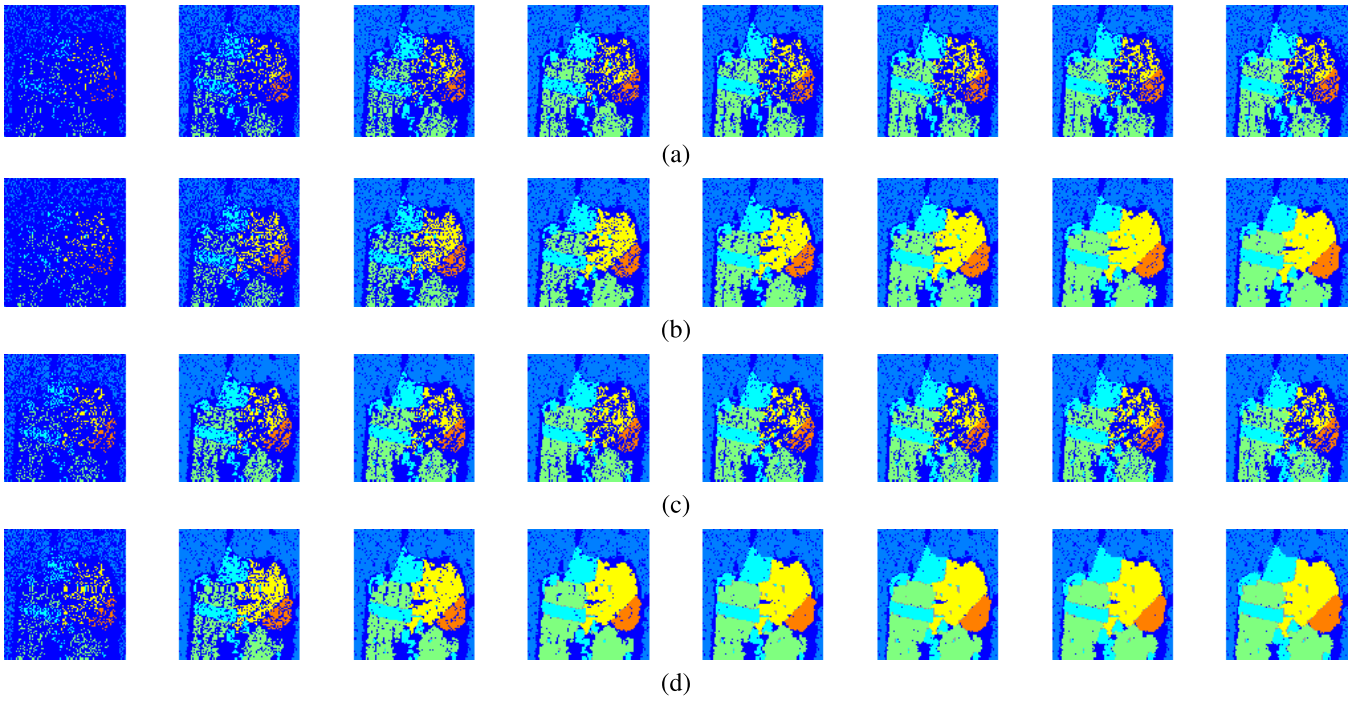


Fig. 7. WLCE samples of San Francisco data. (a) San Francisco : Rule 1, iteration number 8, and window 7 × 7. (b) San Francisco : Rule 2, iteration number 8, and window 7 × 7. (c) San Francisco : Rule 1, iteration number 8, and window 11 × 11. (d) San Francisco : Rule 2, iteration number 8, and window 11 × 11.

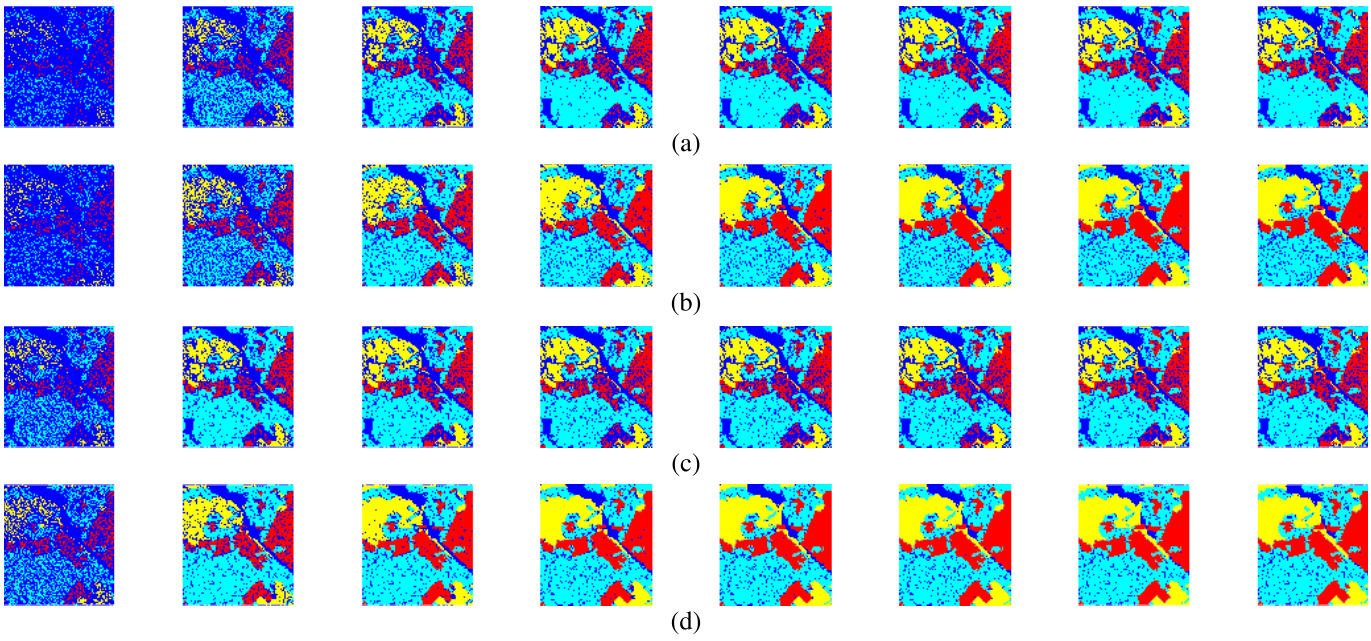


Fig. 8. WLCE samples of Oberpfaffenhofen data. (a) Oberpfaffenhofen : rule 1, iteration number 8, window 7 × 7. (b) Oberpfaffenhofen : rule 2, iteration number 8, window 7 × 7. (c) Oberpfaffenhofen : rule 1, iteration number 8, window 11 × 11. (d) Oberpfaffenhofen : rule 2, iteration number 8, window 11 × 11.

C. Use Wishart Locally Constrained Expansion to Train Convolution Neural Network

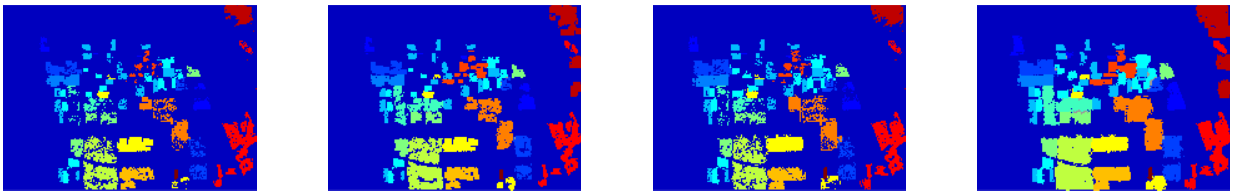
After displaying the above WLCE results, we use WLCE to train CNN to evaluate the proposed algorithm.

In order to show the generality of the algorithm, two different PolSAR data are used. The Flevoland and Weihe Revier datasets—one stands for more classes and detailed, and the other stands for more confusion results in the current literature—are used to show the performance following the overall process of the proposed algorithm.

In this section, we show the CNN primary results that use WLCE samples.

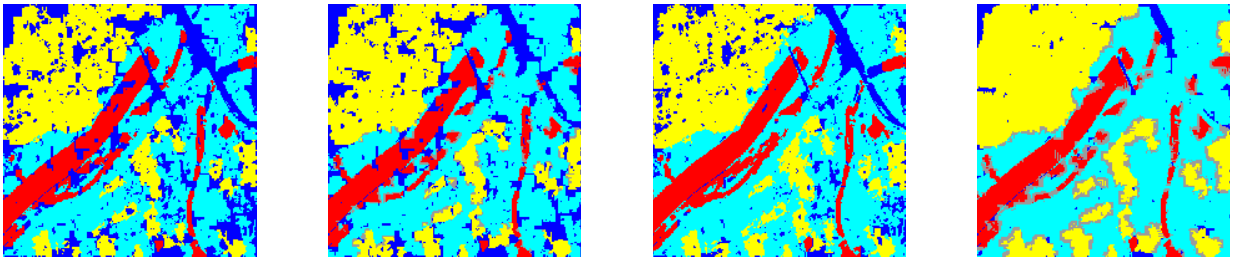
Since this part mainly focuses on the effectiveness evaluation of expansion samples and the influence of different WLCE parameter settings on the experiment results, the MRF smooth coefficient is set as a fixed value in this section, and the window size is set as 7 and 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 9. WLCE samples of Flevoland data. (a) Rule 1, window 7 × 7. (b) Rule 2, window 7 × 7. (c) Rule 1, window 11 × 11. (d) Rule 2, window 11 × 11.



(a) (b) (c) (d)

Fig. 10. WLCE samples of Weihe River data. (a) Rule 1, window 7 × 7. (b) Rule 2, window 7 × 7. (c) Rule 1, window 11 × 11. (d) Rule 2, window 11 × 11.

The results of the Flevoland and Weihe River datasets are shown in Figs. 11 and 12, respectively. The two figures show the changing curves of overall accuracy (OA) coefficients obtained by different WLCE settings when the iteration number increases from 1 to 8, omitting the iteration number from 6 to 7 in order to reduce redundant work. The first curve of w7 + MRF shows the OA of CNN classification with MRF, and the window size of WLCE is 7. The second curve of w7 shows the OA of CNN classification without MRF, and the window size of WLCE is 7. It is the same with the first curve of w5 + MRF and the second curve of w5. The line of CNN shows the CNN classification results using only the original training data without MRF postprocessing. The line of CNNMRF shows the CNN classification results using only the original training data with MRF postprocessing. They are shown here as baselines.

For the Flevoland dataset, it can be seen that CNN with WLCE can improve the performance of the classification, and only one iteration can bring a significant improvement whether CNN is trained with WLCE samples based on Rule 1 or Rule 2. Rule 2 demonstrates better results. In Fig. 11(a), OA has a trend that increases first with the number of iterations and then reaches a stable value. In Fig. 11(b), OA has a trend that increases with the number of iterations continuously. In Fig. 11(b), the lowest classification accuracy with MRF and the lowest classification accuracy without MRF are obviously higher than CNN with original training samples. The results are satisfactory.

For the Weihe River dataset, it can be seen that there is something different, mainly for Rule 1, in Fig. 12(a). Compared with classification results from CNN trained with original samples, classification results from CNN trained with WLCE samples still exhibit higher accuracy, but there is no advantage compared with classification results with MRF. Rule 2 still demonstrates better results with WLCE samples as the training set. In Fig. 12(b), the lowest classification accuracy with MRF and the lowest classification accuracy without MRF are higher than CNN with original training samples. Except for Fig. 12(a), when the number of iterations is 4 and the window size is 7, the classification performance is already satisfactory. In order to study the performance of the proposed algorithm, subsequent experiments are uniformly set with an iteration number of 4 and a window size of 7.

D. Fine-Tune Convolution Neural Network

The results of the above experiments indicate that satisfactory classification performance can be achieved using CNN trained with WLCE samples. Using consistency samples and original training samples to fine-tune the initial trained CNN can make the model closer to the true distribution. There are two groups of the Weihe River dataset experiment shown in the following (see Figs. 13 and 14) to display the fine-tuning effect. WLCE uses a sample-centered rule, the iteration number is set as 4, the window size is set as 7, and the fine-tune epoch is 80.

After CNN fine-tuning, samples of the same type become more concentrated (Group 1, Group 2: a versus b, c versus d), and larger edge contours of the image become smoother (Group 2: a versus b, c versus d). Some lost details reappear due to the enhanced discriminability of the classifier (Group 2: a versus b, c versus d). At the same time, the internal consistence of the region brought about by the large region prior, in the confrontation with sample distribution, shows a decrease.

E. Performance Evaluation

We have made a detailed analysis of each part of the algorithm. Here, we show the overall performance of the algorithm. Wishart, WishartMRF, SVM, SVMMRF, CNN, SPCNN [58], SPGCNN [58], CNN-AFA [56], and CNN-AFAM [56] are used for comparison algorithms. Wishart, WishartMRF, SVM, and SVMMRF are traditional supervised methods, CNN is the benchmark deep learning

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

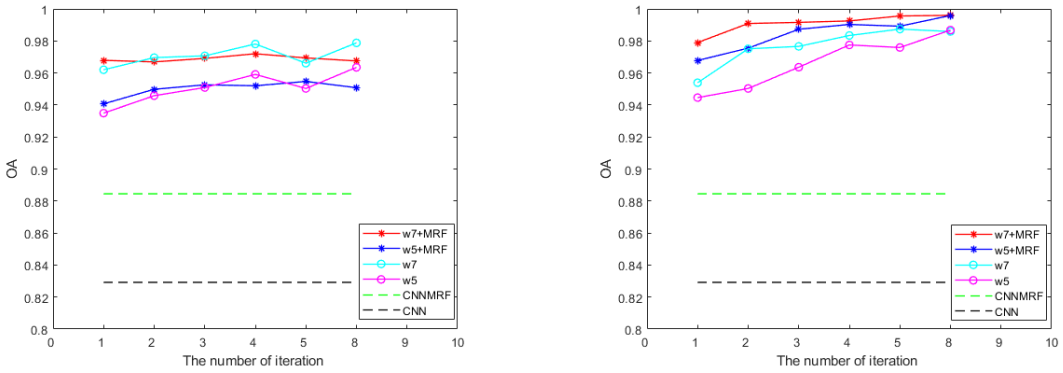


Fig. 11. Classification results of the Flevoland dataset with different WLCE settings when the iteration times increase in the range of 1, 2, 3, 4, 5, 8. (a) Rule 1. (b) Rule 2.

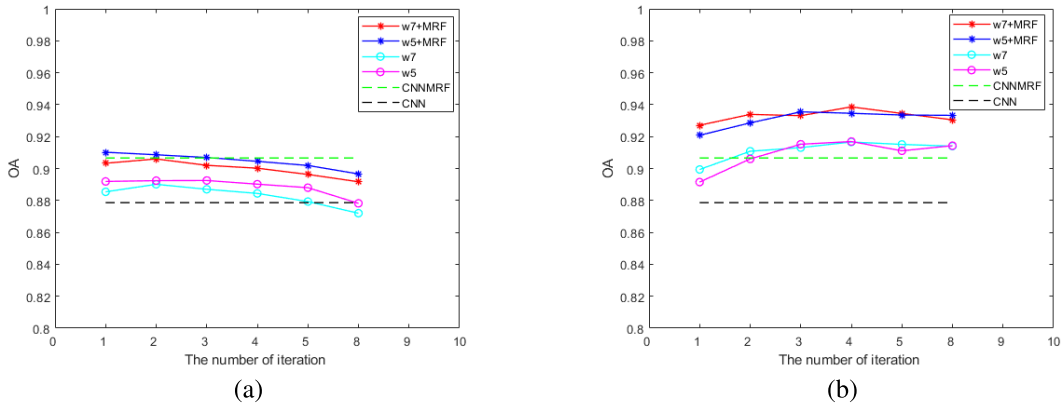


Fig. 12. Classification results of the Weihe River dataset with different WLCE settings when the iteration times increase in the range of 1, 2, 3, 4, 5, 8. (a) Rule 1. (b) Rule 2.

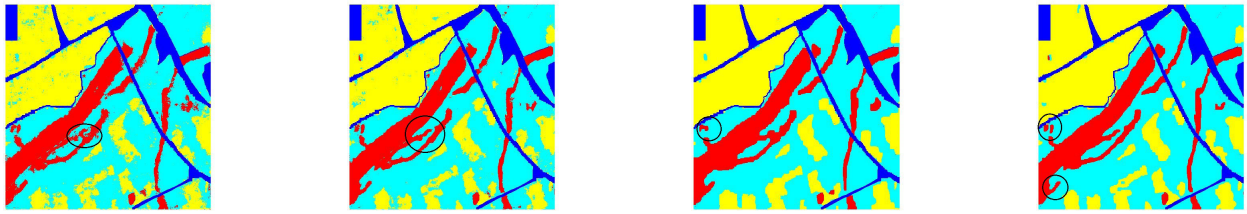


Fig. 13. Group 1: comparison before and after CNN fine-tuning in the Weihe River dataset. (a) First stage of CNN, without MRF. (b) Fine-tuning stage of CNN, without MRF. (c) First stage of CNN, with MRF. (d) Fine-tuning stage of CNN, with MRF.

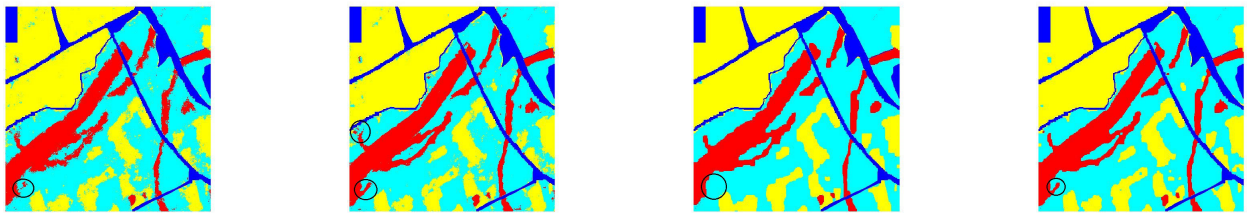


Fig. 14. Group 2: comparison before and after CNN fine-tuning in the Weihe River dataset. (a) First stage of CNN, without MRF. (b) Fine-tuning stage of CNN, without MRF. (c) First stage of CNN, with MRF. (d) Fine-tuning stage of CNN, with MRF.

method, and SPCNN, SPGCNN, CNN-AFA, and CNN-AFAM are semisupervised methods. SVMMRF is the method incorporating SVM and MRF optimization. CNNMRF is the method incorporating CNN and MRF optimization. CNN-AFA is the method incorporating active learning samples and CNN. CNN-AFAM is the method incorporating CNN-AFA and MRF optimization. SPCNN is the method incorporating superpixel expansion samples and CNN. SPGCNN is the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION RESULTS OF FLEVOLAND DATA WITH DIFFERENT METHODS

	Wishart	WishartMRF	SVM	SVMMRF	CNN	CNNMRF	SPCNN
OA	0.7255	0.8196	0.7934	0.8917	0.8118	0.9218	0.9522
stembean	0.7068	0.7753	0.6955	0.8304	0.8050	0.9374	0.9784
rapeseed	0.5419	0.6032	0.5711	0.7565	0.7448	0.9116	0.9734
bare soil	0.8976	0.9871	0.8867	0.9820	0.1642	0.0648	0.9847
potato	0.8544	0.9799	0.8596	0.9928	0.8895	0.9964	0.9634
beet	0.9076	0.9919	0.8590	0.9424	0.8071	0.9299	0.9742
wheat 1	0.6776	0.7906	0.4206	0.3624	0.6178	0.8353	0.8955
peas	0.7568	0.9836	0.8924	0.9664	0.8643	0.9962	0.9707
wheat 3	0.7036	0.8298	0.8824	0.9538	0.9182	0.9885	0.9861
lucerne	0.8988	0.9794	0.8693	0.9991	0.9151	0.9833	0.8539
barley	0.9365	0.9762	0.9173	0.9756	0.9336	1.0000	0.9544
wheat 2	0.6778	0.8142	0.8343	0.9603	0.8165	0.9820	0.9550
grass	0.1347	0.0986	0.4872	0.6282	0.4075	0.6002	0.7410
forest	0.8966	0.9889	0.8459	0.9859	0.8760	0.9866	0.9911
water	0.4992	0.5089	0.9555	1.0000	0.9684	0.9903	0.9965
building	0.8354	0.8803	0.0041	0.0000	0.7306	0.7823	0.6871
	SPGCNN	CNN-AFA	CNN-AFAM	WCCN1	WCCNM1	WCCN2	WCCNM2
OA	0.9772	0.9358	0.9794	0.9622	0.9755	0.9860	0.9943
stembean	0.9841	0.9553	0.9834	0.9987	0.9998	0.9896	0.9997
rapeseed	0.9938	0.8948	0.9639	0.9275	0.9499	0.9820	0.9947
bare soil	0.9959	0.9873	0.9996	0.9996	1.0000	0.9973	1.0000
potato	0.9949	0.9618	0.9923	0.9912	0.9963	0.9809	0.9964
beet	0.9900	0.9391	0.9835	0.9953	1.0000	0.9985	1.0000
wheat 1	0.9724	0.8974	0.9802	0.9201	0.9364	0.9892	1.0000
peas	0.9876	0.9668	0.9907	0.9913	0.9952	0.9939	0.9996
wheat 3	0.9990	0.9398	0.9871	0.9837	0.9960	0.9949	0.9990
lucerne	0.8911	0.9557	0.9696	0.9708	0.9727	0.9636	0.9670
barley	0.9888	0.9409	0.9909	0.9938	0.9975	0.9971	1.0000
wheat 2	0.9909	0.9370	0.9919	0.9963	0.9999	0.9863	0.9990
grass	0.8025	0.8018	0.9209	0.7866	0.8337	0.9177	0.9578
forest	0.9996	0.9329	0.9617	0.9734	0.9896	0.9973	0.9998
water	0.9978	0.9871	0.9985	0.8993	0.9406	0.9970	0.9983
building	0.7156	0.7143	0.7619	0.8966	0.9224	0.8503	0.8476

TABLE II
CLASSIFICATION RESULTS OF WEIHE RIVER WITH DIFFERENT METHODS

	Wishart	WishartMRF	SVM	SVMMRF	CNN	CNNMRF	SPCNN
OA	0.8428	0.8933	0.8621	0.8997	0.8778	0.9071	0.8899
Grass	0.8280	0.8768	0.9158	0.9523	0.8793	0.9106	0.9131
City	0.8181	0.8916	0.7852	0.8436	0.8724	0.9111	0.8709
Water	0.9454	0.9505	0.8566	0.8523	0.8849	0.8869	0.8562
	SPGCNN	CNN-AFA	CNN-AFAM	WCCN1	WCCNM1	WCCN2	WCCNM2
OA	0.9225	0.8798	0.9009	0.8921	0.9193	0.9267	0.9383
Grass	0.9542	0.8640	0.8863	0.9116	0.9397	0.9466	0.9571
City	0.9083	0.8773	0.9071	0.8720	0.9064	0.9041	0.9226
Water	0.8509	0.9367	0.9348	0.8733	0.8814	0.9117	0.9119

method incorporating SPCNN and the graph model. CNN that uses WLCE samples is noted as WCCN. WCCNM is our whole algorithm incorporating WCNN and MRF optimization. 1 represents cluster centered rule expansion. 2 represents sample centered rule expansion.

1) Performance of 0.5% Training Samples: Since this article aims to address the issue of a small number of training samples, we conduct the experiments with 0.5% randomly selected labeled pixels as training samples. The results are shown in Tables I and II and Figs. 15 and 16.

For Flevoland data, from Table I and Fig. 15, we can see that the OA of the proposed WCCNM2 algorithm is 99.43%, which significantly outperforms Wishart, WishartMRF, SVM, SVMMRF, SPCNN, SPGCNN, CNN-AFA, and CNN-AFAM. The classification result is very satisfactory. The accuracy of each class of the proposed WCCNM is also better than other algorithms, only worse for very few classes. As we noted before, WCCN2 and WCCNM2 are also better than WCCN1 and WCCNM1, respectively, especially in rapeseed, wheat 1, grass, and water, but worse in building, lucerne, and slightly worse in stembean and wheat 2.

Without spatial features, the classification results (as shown in Fig. 15) of Wishart and SVM are full of chaos. Without MRF, their OA is below 80%. Due to the addition of spatial relationship information, MRF and CNN are better than single-pixel processing algorithms.

With a CNN classifier, SPCNN, WCCN1, and WCCN2 using neighborhood samples for training obtain more consistent results than CNN and CNN-AFA.

For the three types of training samples' expansion algorithms, superpixel expansion algorithm (SPCNN), active learning expansion algorithm (CNN-AFA), and our proposed WLCE expansion algorithm (WCCN1 and WCCN2), it can be seen that, with the WLCE sample-centered expansion algorithm, the internal region is more consistent. The reason

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

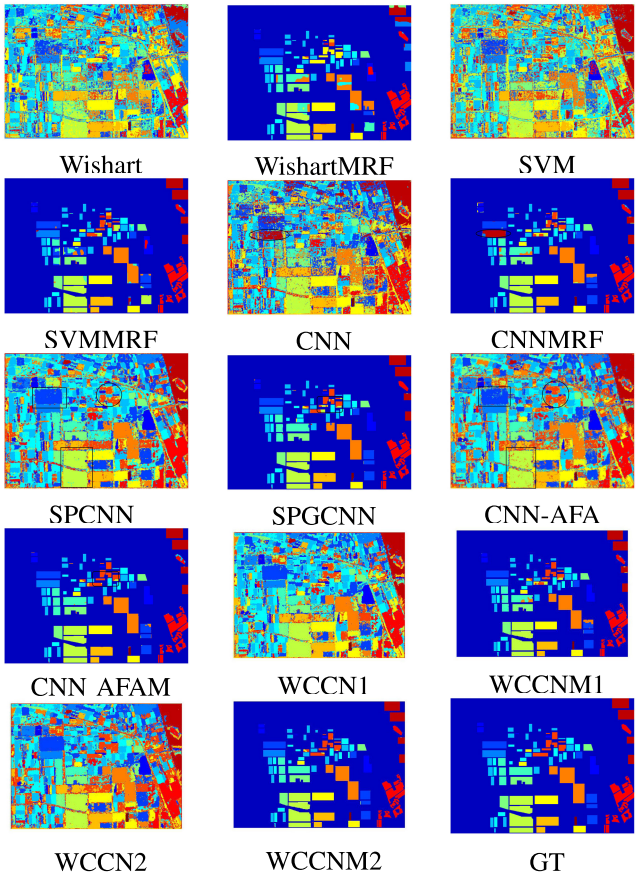


Fig. 15. Classification results of Flevoland data with different methods.

should be that the proposed algorithm can afford more region pixels than the other two that contribute to CNN training.

CNN-AFA and WCCN2 work better in grass region than SPCNN and WCCN1, which is mainly because, when the features provided are difficult to distinguish or the inhomogeneity of pixels is strong, active learning can improve distinguishability, and sample-centered rule can embrace more informative samples.

For Weihe River data, the whole situation is similar to that of Flevoland data. The OA of the proposed WCCNM2 algorithm is 93.83%, which outperforms Wishart, WishartMRF, SVM, SVMMRF, SPCNN, SPGCNN, CNN-AFA, and CNN-AFAM. The accuracy of each class of the proposed WCCNM algorithm is also better than most of the other algorithms. The classification result is satisfactory. The accuracy of each class of the proposed WCCNM2 algorithm is also better than most of the other algorithms, only worse of very small probability. Like Flevoland Data, WCCN2 and WCCNM2 are also, respectively, better than WCCN1 and WCCNM1. Without spatial features, the classification results (as shown in Fig. 16) of the Wishart algorithm and the SVM algorithm are full of chaos. MRF and CNN algorithms are better than single-pixel processing algorithms.

Due to the addition of samples that are easy to distinguish and the increase in the number of samples, CNN-AFA and WCCN2 work better (considering the result after MRF optimization) in the circled water region than CNN and SPCCN. CNN-AFA and WCCN2 work better (considering

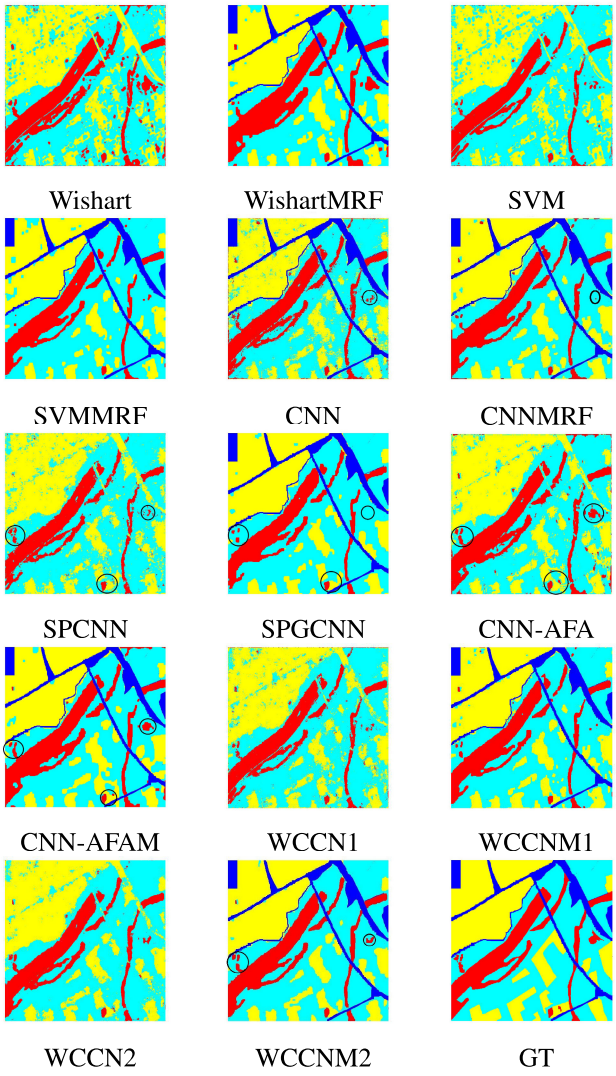


Fig. 16. Classification results of Weihe River data with different methods.

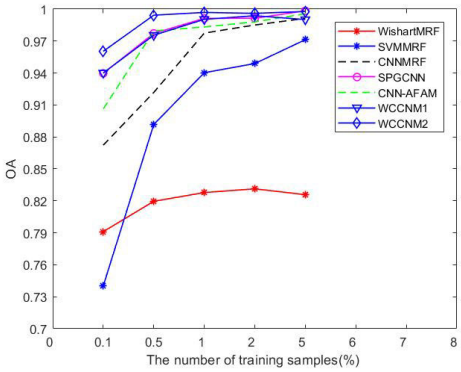


Fig. 17. Classification results of the Flevoland dataset with different training sample rates.

the result after MRF optimization) in regions with small size details than SPCNN and WCCN1, which can be seen in the circled water area. This benefits from a strong discriminability sample expanded by active learning and informative samples expanded by the WLCE sample-centered rule. A very noteworthy phenomenon is that the regional consistency of WCCN2 is higher than that of CNN, CNN-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

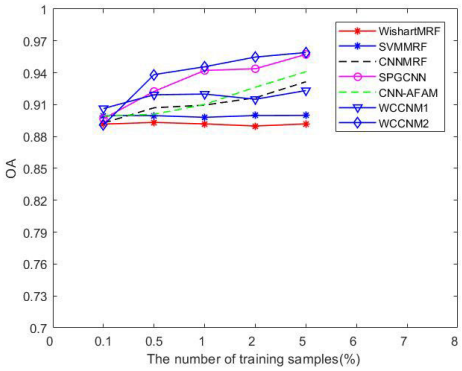


Fig. 18. Classification results of the Weihe River dataset with different training sample rates.

AFA, SPCCN, and WCCN1, indicating the importance of expanding the sample or sample selection strategy.

2) Performance of Different Training Sample Rates: To comprehensively investigate our algorithm, we conduct experiments with five different training sample rates. The results of Flevoland data and Weihe River data are shown in Figs. 17 and 18, respectively. The algorithms' results discussed here are postprocessed by MRF since we aim to discuss the issue of final results.

From the results of the Flevoland data in Fig. 17, it can be seen that WCCNM2 outperforms other algorithms in all cases, and WCCNM1 also achieves good performance. As the number of samples increases, except for WishartMRF, the OAs of the other algorithms significantly increase in most cases. The OAs of WishartMRF and SVMMRF algorithms are lower than CNN algorithms. For CNN algorithms, adding more spatial information helps the algorithm work better (WCCNM2 works better than SPGCNN and WCCNM1, and SPGCNN works better than CNN). It can be seen that, as the number of samples increases, the OAs of SPGCNN and WCCNM2 gradually approach. The reason should be that the increase in training samples makes the expanded samples approximate the full-covering state with high accuracy, which is the expansion limit.

From the results of the Weihe River data in Fig. 18, it can be seen that a similar situation arises. Except when the sampling rate is 0.1%, WCCNM2 performs the best. This exception indicates that, when the sample size is very small, WCCNM2 still needs further investigation. The performance of WCCNM1 in the Weihe River data is not as good as that of Flevoland Data, but it is somewhat unexpected that, compared with other algorithms, WCCNM1 obtains the best performance when the sampling rate is 0.1%. This indicates that WLCE1 still deserves attention.

V. Conclusion and Future Work

This article proposes a sample expansion algorithm WLCE based on region prior and designs a new PolSAR classification algorithm. Four benchmark datasets are used to validate the efficiency of the proposed WCCNM algorithm. Abundant experiments have been done on every part of the algorithm. Experiments show that WLCE can grow extensive samples with high quality, which are very suitable for deep learning. Though the experiments about the two WLCE rules show that the sample-centered rule algorithm obtains better performance, the comparison between them is meaningful. The analyses among the two WLCE algorithms and other sample expanding algorithms (superpixel and active learning) inspire us not only to pay attention to quantity when designing expansion training samples but also how to design expansion samples with rich information. A two-stage CNN training method is designed in the proposed PolSAR classification algorithm. Compared with other PolSAR classification algorithms, it is convinced that the proposed algorithm is state-of-the-art.

References

[1] W. Zhai, H.-F. Shen, C.-L. Huang, and W.-S. Pei, "Building damage information investigation after earthquake using single post-event PolSAR image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 7338–7341.

[2] J.-S. Lee, K. W. Hoppel, S. A. Mango, and A. R. Miller, "Intensity and phase statistics of multilook polarimetric and interferometric SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 5, pp. 1017–1028, Mar. 1994.

[3] Z. Fang, G. Zhang, Q. Dai, and B. Xue, "PolSAR image classification based on complex-valued convolutional long short-term memory network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[4] Y. Duguay, M. Bernier, E. Lévesque, and F. Domine, "Land cover classification in subarctic regions using fully polarimetric RADARSAT-2 data," *Remote Sens.*, vol. 8, no. 9, p. 697, Aug. 2016.

[5] C. Yonezawa, M. Watanabe, and G. Saito, "Polarimetric decomposition analysis of ALOS PALSAR observation data before and after a landslide event," *Remote Sens.*, vol. 4, no. 8, pp. 2314–2328, Aug. 2012.

[6] J.-S. Lee, M. R. Grunes, T. L. Ainsworth, L.-J. Du, D. L. Schuler, and S. R. Cloude, "Unsupervised classification using polarimetric decomposition and the complex Wishart classifier," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 5, pp. 2249–2258, Mar. 1999.

[7] A. P. Doulgeris, S. N. Anfinsen, and T. Eltoft, "Automated non-Gaussian clustering of polarimetric synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3665–3676, Oct. 2011.

[8] X. Deng, C. López-Martínez, J. Chen, and P. Han, "Statistical modeling of polarimetric SAR data: A survey and challenges," *Remote Sens.*, vol. 9, no. 4, p. 348, Apr. 2017.

[9] C. C. Freitas, A. C. Frery, and A. H. Correia, "The polarimetric $\mathcal{G}$ distribution for SAR data analysis," *Environmetrics*, vol. 16, no. 1, pp. 13–31, 2005.

[10] W. Song, M. Li, P. Zhang, Y. Wu, L. Jia, and L. An, "The WGΓ distribution for multilook polarimetric SAR data and its application," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 10, pp. 2056–2060, Oct. 2015.

[11] S.-W. Chen and C.-S. Tao, "PolSAR image classification using polarimetric-feature-driven deep convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 627–631, Apr. 2018.

[12] L. Jiao and F. Liu, "Wishart deep stacking network for fast POLSAR image classification," *IEEE Trans. Image Process.*, vol. 25, no. 7, pp. 3273–3286, Jul. 2016.

[13] S. R. Cloude and E. Pottier, "A review of target decomposition theorems in radar polarimetry," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 2, pp. 498–518, Mar. 1996.

[14] S. R. Cloude and E. Pottier, "An entropy based classification scheme for land applications of polarimetric SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 1, pp. 68–78, Feb. 1997.

[15] A. Freeman and S. L. Durden, "A three-component scattering model for polarimetric SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 3, pp. 963–973, May 1998.

[16] K.-U. Khan, J. Yang, and W. Zhang, "Unsupervised classification of polarimetric SAR images by EM algorithm," *IEICE Trans. Commun.*, vol. 90, no. 12, pp. 3632–3642, Dec. 2007.

[17] X. She, J. Yang, and W. Zhang, "The boosting algorithm with application to polarimetric SAR image classification," in *Proc. 1st Asian Pacific Conf. Synth. Aperture Radar*, 2007, pp. 779–783.

[18] E. Krogager, "New decomposition of the radar target scattering matrix," *Electron. Lett.*, vol. 26, no. 18, pp. 1525–1527, Aug. 1990.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.

[19] J. R. Huynen, "Phenomenological theory of radar targets," in *Electromagnetic Scattering*, 1978, pp. 653–712.

[20] J. S. Lee and M. R. Grunes, "Classification of multi-look polarimetric SAR data based on complex Wishart distribution," in *Proc. Nat. Telesystems Conf.*, 1992, pp. 7–21.

[21] J. S. Lee, D. L. Schuler, R. H. Lang, and K. J. Ranson, "K-distribution for multi-look processed polarimetric SAR imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Mar. 1994, pp. 2179–2181.

[22] S. Khan and R. Guida, "Application of Mellin-kind statistics to polarimetric $\mathcal{G}$ distribution for SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3513–3528, Jun. 2014.

[23] L. Bombrun and J.-M. Beaulieu, "Fisher distribution for texture modeling of polarimetric SAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 3, pp. 512–516, Jul. 2008.

[24] Y. Wu, K. Ji, W. Yu, and Y. Su, "Region-based classification of polarimetric SAR images using Wishart MRF," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 668–672, Oct. 2008.

[25] T. M. Nguyen and Q. M. J. Wu, "Gaussian-mixture-model-based spatial neighborhood relationships for pixel labeling problem," *IEEE Trans. Syst., Man, Cybern., B, Cybern.*, vol. 42, no. 1, pp. 193–202, Feb. 2012.

[26] C. Liu, W. Liao, H.-C. Li, K. Fu, and W. Philips, "Unsupervised classification of multilook polarimetric SAR data using spatially variant Wishart mixture model with double constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5600–5613, Oct. 2018.

[27] C. Liu, H.-C. Li, W. Liao, W. Philips, and W. Emery, "Variational textured Dirichlet process mixture model with pairwise constraint for unsupervised classification of polarimetric SAR images," *IEEE Trans. Image Process.*, vol. 28, no. 8, pp. 4145–4160, Aug. 2019.

[28] W. Song, M. Li, P. Zhang, Y. Wu, X. Tan, and L. An, "Mixture WG Γ-MRF model for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 905–920, Feb. 2018.

[29] S. Wang et al., "Semi-supervised PolSAR image classification based on improved tri-training with a minimum spanning tree," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8583–8597, Dec. 2020.

[30] M. Hellmann, G. Jager, E. Kratzschmar, and M. Habermeyer, "Classification of full polarimetric SAR-data using artificial neural networks and fuzzy algorithms," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Feb. 1999, pp. 1995–1997.

[31] Y. Zhou, H. Wang, F. Xu, and Y.-Q. Jin, "Polarimetric SAR image classification using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1935–1939, Dec. 2016.

[32] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[33] C. Yang, B. Hou, B. Ren, Y. Hu, and L. Jiao, "CNN-based polarimetric decomposition feature selection for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8796–8812, Nov. 2019.

[34] W. Hua, S. Wang, W. Xie, Y. Guo, and X. Jin, "Dual-channel convolutional neural network for polarimetric SAR images classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3201–3204.

[35] Y. Cui et al., "Polarimetric multipath convolutional neural network for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5207118.

[36] R. Yang, X. Xu, Z. Xu, H. Dong, R. Gui, and F. Pu, "Dynamic fractal texture analysis for PolSAR land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5991–6002, Aug. 2019.

[37] X. Zhang, K. Z. Deng, H. D. Fan, and H. Zhao, "PolSAR SVM supervised classification method combining with polarimetric target decomposition," *Appl. Res. Comput.*, vol. 30, no. 1, pp. 1–12, 2013.

[38] W. Xie et al., "POLSAR image classification via Wishart-AE model or Wishart-CAE model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3604–3615, Aug. 2017.

[39] J. Wang, B. Hou, L. Jiao, and S. Wang, "POL-SAR image classification based on modified stacked autoencoder network and data distribution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1678–1695, Mar. 2020.

[40] Q. Wu, B. Hou, Z. Wen, and L. Jiao, "Variational learning of mixture Wishart model for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 141–154, Jan. 2019.

[41] T. Gadhiya and A. K. Roy, "Superpixel-driven optimized Wishart network for fast PolSAR image classification using global k-means algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 97–109, Jan. 2020.

[42] B. Ren, Y. Zhao, B. Hou, J. Chanussot, and L. Jiao, "A mutual information-based self-supervised learning model for PolSAR land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9224–9237, Nov. 2021.

[43] J. Geng, H. Wang, J. Fan, and X. Ma, "SAR image classification via deep recurrent encoding neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2255–2269, Apr. 2018.

[44] L. Wang, X. Xu, H. Dong, R. Gui, R. Yang, and F. Pu, "Exploring convolutional LSTM for PolSAR image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 8452–8455.

[45] K. Huang, W. Nie, and N. Luo, "Fully polarized SAR imagery classification based on deep reinforcement learning method using multiple polarimetric features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 3719–3730, Oct. 2019.

[46] L. Zhang, W. Ma, and D. Zhang, "Stacked sparse autoencoder in PolSAR data classification using local spatial information," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 9, pp. 1359–1363, Sep. 2016.

[47] J. Ni, F. Zhang, Q. Yin, Y. Zhou, H.-C. Li, and W. Hong, "Random neighbor pixel-block-based deep recurrent learning for polarimetric SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7557–7569, Sep. 2021.

[48] W. Xie, G. Ma, F. Zhao, H. Liu, and L. Zhang, "PolSAR image classification via a novel semi-supervised recurrent complex-valued convolution neural network," *Neurocomputing*, vol. 388, pp. 255–268, May 2020.

[49] D. Li, Y. Gu, S. Gou, and L. Jiao, "Full polarization SAR image classification using deep learning with shallow feature," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 4566–4569.

[50] H. Bi, J. Sun, and Z. Xu, "A graph-based semisupervised deep learning model for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2116–2132, Apr. 2019.

[51] J. Geng, X. Ma, J. Fan, and H. Wang, "Semisupervised classification of polarimetric SAR image via superpixel restrained deep neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 122–126, Jan. 2018.

[52] H. Liu, R. Luo, F. Shang, X. Meng, S. Gou, and B. Hou, "Semi-supervised deep metric learning networks for classification of polarimetric SAR data," *Remote Sens.*, vol. 12, no. 10, p. 1593, May 2020.

[53] Z. Fang, G. Zhang, Q. Dai, Y. Kong, and P. Wang, "Semisupervised deep convolutional neural networks using pseudo labels for PolSAR image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[54] Y. Li et al., "Semi-supervised PolSAR image classification based on self-training and superpixels," *Remote Sens.*, vol. 11, no. 16, p. 1933, Aug. 2019.

[55] W. Hua, S. Wang, H. Liu, K. Liu, Y. Guo, and L. Jiao, "Semisupervised PolSAR image classification based on improved cotraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 4971–4986, Nov. 2017.

[56] H. Bi et al., "An active deep learning approach for minimally-supervised PolSAR image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3185–3188.

[57] J. Nazarinezhad and M. Dehghani, "A contextual-based segmentation of compact PolSAR images using Markov random field (MRF) model," *Int. J. Remote Sens.*, vol. 40, no. 3, pp. 985–1010, Feb. 2019.

[58] B. Hou, Q. Wu, Z. Wen, and L. Jiao, "Robust semisupervised classification for PolSAR image with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6440–6455, Nov. 2017.



Jing Chen received the B.S. degree from Central South University, Changsha, China, in 2002, and the M.S. degree from Henan University, Kaifeng, China, in 2009. She is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University, Xi'an, China.

She is currently a Lecturer with Henan University. Her research interests include pattern recognition, image classification, and machine learning in remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.



Biao Hou (Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University, where he is currently a Professor. His research interests include compressive sensing, deep learning, and synthetic aperture radar image interpretation.



Bo Ren (Member, IEEE) received the B.S. degree in telecommunications engineering from Northwest University, Xi'an, China, in 2011, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2017.

Since 2018, he has been an Assistant Professor with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. His research interests include synthetic aperture radar image interpretation, and sparse representation and machine learning in remote sensing images.



Qian Wu received the B.S. and Ph.D. degrees from the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University, Xi'an, China, in 2013 and 2020, respectively.

She is currently a Lecturer with the Institute of Information and Navigation, Air Force Engineering University, Xi'an. Her research interests include Bayesian learning, deep learning, and intelligent navigation.



Licheng Jiao (Fellow, IEEE) was born in Shaanxi, China, in 1959. He received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an. He is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. His research interests include signal and image processing, nonlinear circuits and systems theory, learning theory and algorithms, optimization problems, wavelet theory, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:40:26 UTC from IEEE Xplore. Restrictions apply.